

Do Explainers Agree on Which Features Matter?

Instance-Level Agreement between SHAP and LIME on Three Tabular Datasets

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Abstract. Post-hoc explainers are often used as if they were interchangeable, yet two explainers may name different features for the same prediction. This study measures how much SHAP and LIME agree at the level of the single instance. Both methods explained the same instances of the same models on three tabular datasets (Adult, German Credit and Breast Cancer Wisconsin), giving 31,411 paired explanations in 86 runs. The analysis plan was registered before any agreement statistic was computed; runs, not instances, are the unit of summary, and intervals resample random seeds. The mean overlap of the five most important features (Jaccard index) was 0.393 on Adult, 0.387 on German Credit and 0.714 on Breast Cancer, well above the overlap of random feature sets but far from identity, and the ordering of the features agreed weakly on two of the three datasets. Agreement depended on the model family and did not depend on the number of explained instances. It was not consistently lower for misclassified instances. Disagreement was larger where the local fidelity of LIME was lower. A post hoc analysis shows that the agreement of signs between the two methods is driven by the predicted class, because LIME without discretisation returns local slopes while SHAP returns signed contributions; sign agreement between them should not be read as a measure of disagreement.

Keywords. Explainable artificial intelligence, post-hoc explanations, disagreement problem, feature attribution, SHAP, LIME.

1 Introduction

Post-hoc explanation methods describe a prediction of a trained model in terms of its input features.

SHAP [16] and LIME [19] are the most used for tabular data: each returns, for one instance, a signed weight per feature. In practice the two are treated as alternatives, and the choice between them is made on cost or habit. That practice assumes that the explanation a user sees does not depend much on the method.

The assumption has been questioned. Krishna et al. [14] named the *disagreement problem*: explanation methods often identify different top features for the same prediction, and practitioners have no principled way to resolve the conflict. Similar observations were made for text models [18] and for defect prediction [21]. If the features shown to a user change with the explainer, an explanation is a property of the pair (model, explainer), not of the model alone, and conclusions drawn from it need the same caution [6, 22].

Most evidence on disagreement comes from a small number of models per dataset and treats the explained instances as independent observations. Two points remain open for tabular classifiers. The first is how much agreement varies with the model family, the dataset and the number of explained instances when the same instances are explained by both methods. The second is whether disagreement is related to two properties that practitioners care about: whether the prediction is correct, and how good each explanation is by the usual quality measures.

This study addresses both points with instance-paired explanations from an existing benchmark co-

hort and a new, instance-aligned cohort produced for this purpose. The research questions are:

- **RQ1.** For the same instance and model, how much do SHAP and LIME agree on the most influential features and on their order?
- **RQ2.** How does this agreement vary with the model family, the sampling intensity and the dataset?
- **RQ3.** Is agreement different for correctly classified and misclassified instances?
- **RQ4.** Is disagreement on an instance associated with the fidelity or the stability of either explanation?

A secondary, descriptive question is how the features used by two explainers of a different kind, Anchors [20] and DiCE [17], overlap with those of SHAP and LIME.

The contributions are: (i) estimates of SHAP–LIME agreement on 31,411 paired explanations across five model families and three datasets, with the run as the unit of summary and intervals clustered by random seed; (ii) evidence that agreement depends on the model family and the dataset, not on the number of explained instances, and is not consistently related to prediction correctness; (iii) an association between disagreement and the local fidelity of LIME; and (iv) a methodological finding: the sign agreement between SHAP and LIME without discretisation is determined mostly by the predicted class, because the two methods return different quantities.

2 Related Work

Disagreement between explainers. Krishna et al. [14] formalised disagreement with measures of feature agreement, rank agreement, sign agreement and rank correlation over the top- k features, and reported frequent disagreement across datasets and models, together with interviews showing that practitioners resolve it with ad hoc rules. Neely et al. [18] found low rank correlation between attribution methods for text classifiers and argued that agreement should not be used to validate a

method. Roy et al. [21] reported disagreement between LIME and SHAP for defect prediction models, on the features and on their sign.

Why methods differ. Han et al. [9] showed that several attribution methods are local function approximations of the model that differ in the neighbourhood and the loss they use, so that no method is best for every neighbourhood. Garreau and von Luxburg [8] analysed LIME for tabular data and showed that, for a linear model, its coefficients are proportional to the gradient of the model. SHAP values, in contrast, distribute the difference between the prediction and an expected prediction among the features [15, 16]. This difference of meaning matters for the sign analysis in Section 4.3.

Reliability of explanations. Explanations can change under small perturbations of the input [1] and can be manipulated [23]. Stability and fidelity are therefore common quality criteria. This study relates them to disagreement at the level of the instance, which, to our knowledge, earlier work on tabular data has not reported with a clustered design.

3 Materials and Methods

3.1 Datasets, Models and Runs

Three public binary classification datasets were used: Adult [2] (108 encoded features), German Credit [11] (61 encoded features) and Breast Cancer Wisconsin, diagnostic [24] (30 numeric features). Numeric features were standardised and categorical features were one-hot encoded; each encoded column is one feature in every analysis.

On Adult, five model families were explained: logistic regression (LR), a support vector machine with probability outputs (SVM), a multilayer perceptron (MLP), a random forest (RF) [3] and gradient boosting (XGB) [5]. Each explainer was run with five random seeds and three sampling intensities ($n = 50, 100$ and 200 test instances per quadrant of the confusion matrix), so that correct and incorrect predictions are both represented. On German Credit and Breast Cancer, RF and XGB models were trained with three seeds each and

explained with one intensity (up to 100 instances per quadrant).

The Adult runs and the SHAP and Anchors runs of the other two datasets belong to a benchmark cohort described in [10]. That work compares explainers by quality measures; no result of it is repeated here. A companion manuscript under review studies explanation quality on correct and incorrect predictions with the same cohort; it does not study agreement between explainers.

The earlier LIME runs on German Credit and Breast Cancer explained a different sample of instances and stored only averages. *A new LIME cohort was therefore produced for this study:* for each dataset, model and seed, LIME explained exactly the instances stored in the corresponding SHAP run. The model binaries were regenerated from the training recipe in the repository, and the run was allowed only after the regenerated model reproduced the stored training metrics, the feature order and the stored label and prediction of every target instance.

3.2 Explainers

SHAP. TreeSHAP [15] was used for RF and XGB, and KernelSHAP [16] with 50 background instances for LR, SVM and MLP. The stored value of a feature is its signed contribution to the output for class 1, relative to the expected output on the background data.

LIME. The tabular explainer [19] was used with 1,000 perturbed samples, kernel width 3.0, ten selected features and *without discretisation of continuous features*. With this setting, the stored value of a feature is the coefficient of the local weighted linear model for class 1, that is, a local slope.

Anchors and DiCE (Adult, secondary analysis). Anchors [20] returns a rule; its feature set is the set of features in the rule. DiCE [17] returns counterfactual instances; its feature set is the set of features whose value changes. Neither is a signed attribution, so only feature sets are compared.

For each instance, a run stores the ten features with the largest absolute value, in that order. All measures below use these stored lists.

3.3 Agreement Measures

Let S_k and L_k be the sets of the k features with the largest absolute SHAP and LIME values for one instance. The primary measure is the top-5 Jaccard overlap [12]:

$$J_5 = \frac{|S_5 \cap L_5|}{|S_5 \cup L_5|}. \quad (1)$$

Three shared features out of five give $J_5 = 3/5 \approx 0.6$, and four give $J_5 = 4/5 \approx 0.8$. The top-10 overlap J_{10} is a sensitivity measure.

Rank concordance is Kendall's τ_b [13] computed on the union of the two top-10 lists; a feature absent from a list receives rank 11, and ties use midranks.

Sign agreement is the fraction of the features in $S_5 \cap L_5$, with both values different from zero, whose SHAP and LIME values have the same sign.

As a reference, not as a threshold, the expected value of J_k for two independent random sets of k features out of p was computed exactly from the hypergeometric distribution. For $k = 5$ it is 0.026 on Adult, 0.047 on German Credit and 0.099 on Breast Cancer.

3.4 Pairing and Quality Control

Explanations were paired by dataset, model, seed, intensity and instance identifier. Only instances with a valid explanation from both methods and the same stored label and prediction entered the analysis. The source runs contained 278 records with an error and no explanation, mostly KernelSHAP on the SVM; these were excluded, and one SVM block had no usable SHAP run. Of the 87 blocks, 76 had identical sets of valid instances for the two methods; in the others the intersection was used, which left 4,936 valid records unpaired. No paired record had a different label or prediction in the two runs. The result is 31,411 paired explanations in 86 runs: 29,667 on Adult (74 runs), 1,060 on German Credit and 684 on Breast Cancer (six runs each). Kendall's τ_b was undefined for 0 pairs and sign agreement for 516 pairs without a shared non-zero feature.

3.5 Statistical Summary

The same test instances recur across models, seeds and intensities, so instances are not independent replicates. Each measure was first averaged within a run (dataset, model, seed, intensity). The point estimate of a group is the mean of its run means. The 95% interval is a percentile bootstrap with 2,000 resamples in which random seeds, not instances, are resampled with replacement, and all runs of a sampled seed are kept together [4, 7]. There are five seeds on Adult and three on the other datasets, so the intervals describe variation between seeds and are not precise.

For RQ3, the difference between the mean J_5 of misclassified and of correctly classified instances was computed within each run that had at least ten instances in each group, averaged over intensities for each model and seed, and summarised by model. For RQ4, the Spearman correlation between disagreement ($1 - J_5$) and each of four quality values of the instance (fidelity and stability of the SHAP and of the LIME explanation) was computed within each run and averaged by seed. Fidelity is the coefficient of determination between the model output and its reconstruction from the explanation weights on a neighbourhood of the instance; stability is the mean cosine similarity between explanations of slightly perturbed copies of the instance. Both were stored with each explanation when the runs were made.

No hypothesis test was planned and no p -value is reported. The estimates for RQ2 to RQ4 are exploratory.

3.6 Registration and Deviations

The analysis plan, with the measures, the unit of analysis, the exclusion rules and the bootstrap settings, was committed to the repository before any agreement statistic was computed and before the new LIME cohort was run. Two refinements made before the analysis are recorded in the plan. One analysis is *post hoc*: after the planned sign agreement was inspected, it was broken down by predicted class and the constancy of the sign of each feature was measured (Section 4.3). It is reported as such.

4 Results

4.1 Agreement on the Top Features (RQ1)

Table 1 and Fig. 1 give the agreement estimates. On Adult, the mean top-5 overlap was 0.393 (95% interval 0.377 to 0.410); on German Credit it was 0.387 (0.343 to 0.426). Both are below the value for three shared features out of five. On Breast Cancer the overlap was 0.714 (0.671 to 0.792), above the value for four shared features. In all three datasets the overlap was several times the expected overlap of random sets.

The order of the features agreed less than their identity. Kendall's τ_b was 0.126 on Adult and 0.122 on German Credit, and 0.517 on Breast Cancer. Extending the lists to ten features raised the overlap on Adult and German Credit (0.507 and 0.480) and lowered it slightly on Breast Cancer (0.669).

4.2 Model Family, Sampling Intensity and Dataset (RQ2)

On Adult, agreement differed between model families. The overlap was highest for logistic regression (0.485) and lowest for the multilayer perceptron (0.335); the intervals of these two families do not overlap. Rank concordance was close to zero for the multilayer perceptron (0.043) and for gradient boosting (0.028, with an interval that includes zero). On German Credit the overlap was higher for the random forest (0.464) than for gradient boosting (0.309); on Breast Cancer the two families were close.

The sampling intensity had no visible effect: pooled over models, the top-5 overlap on Adult was 0.395, 0.394 and 0.390 for 50, 100 and 200 instances per quadrant. Explaining more instances makes the estimate of agreement more precise; it does not change the agreement.

The dataset mattered most. Adult and German Credit, with many one-hot encoded features, gave similar and low agreement; Breast Cancer, with 30 numeric features, gave much higher agreement for the same two model families. Three datasets do not allow this difference to be attributed to a specific property of the data.

Table 1. Agreement between SHAP and LIME: mean of run means and seed-clustered 95% interval. Adult rows pool the three sampling intensities

Dataset	Model	Runs	Pairs	Top-5 Jaccard	Top-10 Jaccard	Kendall τ_b
Adult	LR	15	7,000	0.485 [0.464, 0.505]	0.617 [0.584, 0.640]	0.263 [0.221, 0.304]
	SVM	14	2,064	0.395 [0.388, 0.404]	0.516 [0.489, 0.539]	0.171 [0.148, 0.187]
	MLP	15	7,000	0.335 [0.326, 0.341]	0.441 [0.435, 0.448]	0.043 [0.029, 0.054]
	RF	15	6,603	0.389 [0.360, 0.415]	0.560 [0.530, 0.590]	0.129 [0.090, 0.168]
	XGB	15	7,000	0.363 [0.328, 0.399]	0.404 [0.384, 0.422]	0.028 [−0.020, 0.071]
	All	74	29,667	0.393 [0.377, 0.410]	0.507 [0.487, 0.525]	0.126 [0.100, 0.153]
German Credit	RF	3	523	0.464 [0.450, 0.471]	0.542 [0.519, 0.572]	0.237 [0.212, 0.251]
	XGB	3	537	0.309 [0.235, 0.381]	0.419 [0.395, 0.440]	0.008 [−0.101, 0.096]
	All	6	1,060	0.387 [0.343, 0.426]	0.480 [0.457, 0.506]	0.122 [0.055, 0.172]
Breast Cancer	RF	3	342	0.738 [0.678, 0.790]	0.665 [0.643, 0.691]	0.526 [0.494, 0.571]
	XGB	3	342	0.690 [0.612, 0.795]	0.673 [0.666, 0.681]	0.507 [0.457, 0.551]
	All	6	684	0.714 [0.671, 0.792]	0.669 [0.654, 0.682]	0.517 [0.475, 0.561]

4.3 Agreement of Signs (Post Hoc)

The planned sign agreement on shared top-5 features was 0.717 on Adult, 0.511 on German Credit and 0.374 on Breast Cancer. The last value is below one half, on the dataset where the two methods agree most on which features matter. This led to the post hoc analysis in Table 2 and Fig. 2.

Sign agreement depended strongly on the predicted class. On Breast Cancer it was 0.977 for instances predicted as class 0 and 0.036 for instances predicted as class 1. On Adult it was 0.875 for class 1 and 0.506 for class 0, and the same pattern appeared in four of the five model families; the multilayer perceptron was the exception.

The reason is that the two methods return different quantities (Section 3.2). The LIME coefficient of a continuous feature is a local slope. The slope of a feature with a monotone effect has the same sign for most instances. A SHAP value is a contribution relative to the expected output: for the same feature it is positive for instances on one side of the reference and negative on the other. The two signs therefore coincide for the instances of one predicted class and differ for the other. The constancy measure confirms it: on Breast Cancer, a feature kept its majority sign in 0.993 of its LIME explanations and in 0.621 of its SHAP explanations

Table 2. Sign agreement on shared top-5 features, overall and by predicted class, and constancy of the sign of a feature across instances (share of instances with the majority sign). Post hoc

Dataset	Model	Sign agreement			Constancy	
		All	Cl. 0	Cl. 1	SHAP	LIME
Adult	LR	0.696	0.527	0.864	0.750	0.878
	SVM	0.829	0.332	0.956	0.837	0.896
	MLP	0.689	0.706	0.672	0.755	0.847
	RF	0.696	0.418	0.975	0.704	0.837
	XGB	0.682	0.443	0.916	0.744	0.795
	All	0.717	0.506	0.875	0.757	0.850
German Credit	RF	0.470	0.608	0.430	0.685	0.948
	XGB	0.553	0.716	0.494	0.673	0.877
	All	0.511	0.662	0.462	0.679	0.912
Breast Cancer	RF	0.377	0.975	0.041	0.622	0.998
	XGB	0.371	0.979	0.032	0.621	0.988
	All	0.374	0.977	0.036	0.621	0.993

(0.912 and 0.679 on German Credit; 0.850 and 0.757 on Adult, where most features are one-hot indicators).

Sign agreement between SHAP and LIME without discretisation is thus not a measure of disagreement about the direction of an effect. It measures mostly on which side of the reference the instance lies.

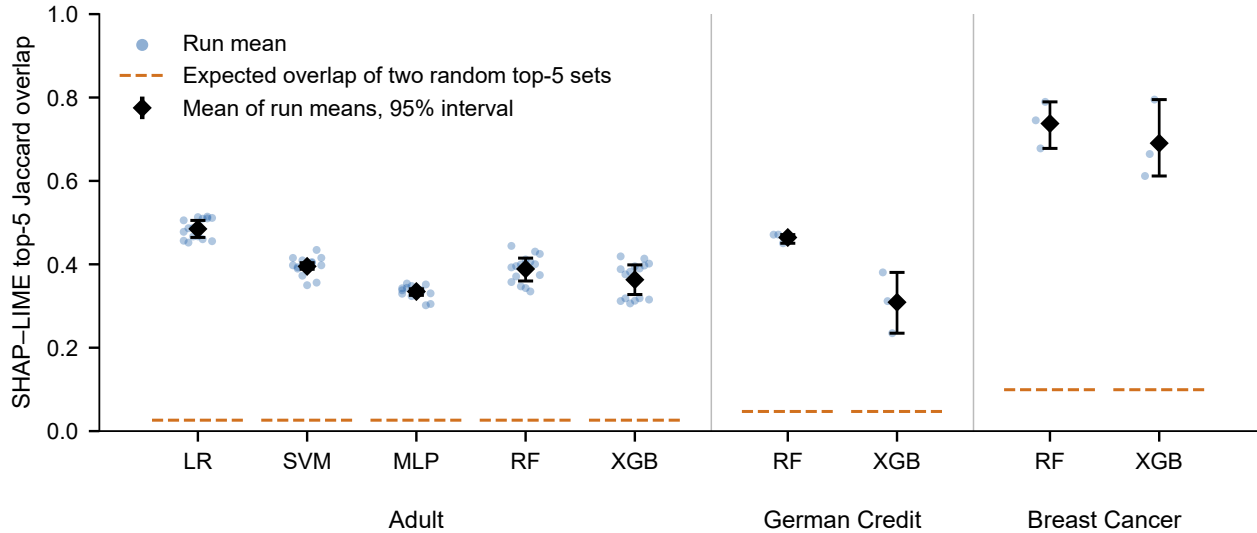


Fig. 1. Top-5 overlap between SHAP and LIME by dataset and model family. Each point is the mean of one run

4.4 Correct and Misclassified Instances (RQ3)

Table 3 and Fig. 3 give the difference in top-5 overlap between misclassified and correctly classified instances. The direction was not consistent. For the multilayer perceptron, agreement was lower on misclassified instances (difference -0.059 , interval -0.069 to -0.049 , same direction in all five seeds). For the random forest it was higher (0.045 , 0.029 to 0.064 , all five seeds), and slightly higher for gradient boosting (0.014). For logistic regression and the support vector machine the difference was near zero. On German Credit the differences were small and negative, with three seeds. On Breast Cancer no run had ten misclassified instances, so the contrast could not be estimated. The largest absolute difference was 0.059 , small compared with the differences between model families.

4.5 Disagreement and Explanation Quality (RQ4)

Table 4 and Fig. 4 give the within-run Spearman correlation between disagreement and the four quality values. The clearest pattern concerns the fidelity of LIME: on Adult the correlation was negative for all five model families (from -0.45 to -0.19), and it was negative on German Credit (-0.26 and -0.24). The two methods disagreed

Table 3. Top-5 overlap for correctly classified (C) and misclassified (M) instances. Diff. is M minus C; Neg. is the number of seeds with a negative difference; GC is German Credit

Data	Model	C	M	Diff.	95% interval	Neg.
Adult	LR	0.486	0.483	-0.003	$[-0.023, 0.012]$	2/5
Adult	SVM	0.388	0.387	-0.001	$[-0.017, 0.020]$	3/4
Adult	MLP	0.364	0.305	-0.059	$[-0.069, -0.049]$	5/5
Adult	RF	0.367	0.412	0.045	$[0.029, 0.064]$	0/5
Adult	XGB	0.356	0.370	0.014	$[0.003, 0.022]$	1/5
GC	RF	0.469	0.452	-0.017	$[-0.036, 0.001]$	2/3
GC	XGB	0.311	0.305	-0.005	$[-0.010, -0.001]$	3/3

more on the instances where the local linear model of LIME reproduced the model less well. The correlations with the fidelity of SHAP were closer to zero and of mixed sign.

For stability there was no common pattern. On Adult the correlations were small and mostly positive, except for the multilayer perceptron. On Breast Cancer with the random forest, disagreement was higher for instances with less stable explanations of either method (-0.56 and -0.54), with three runs. These are associations within runs; they do not show that low fidelity causes disagreement.

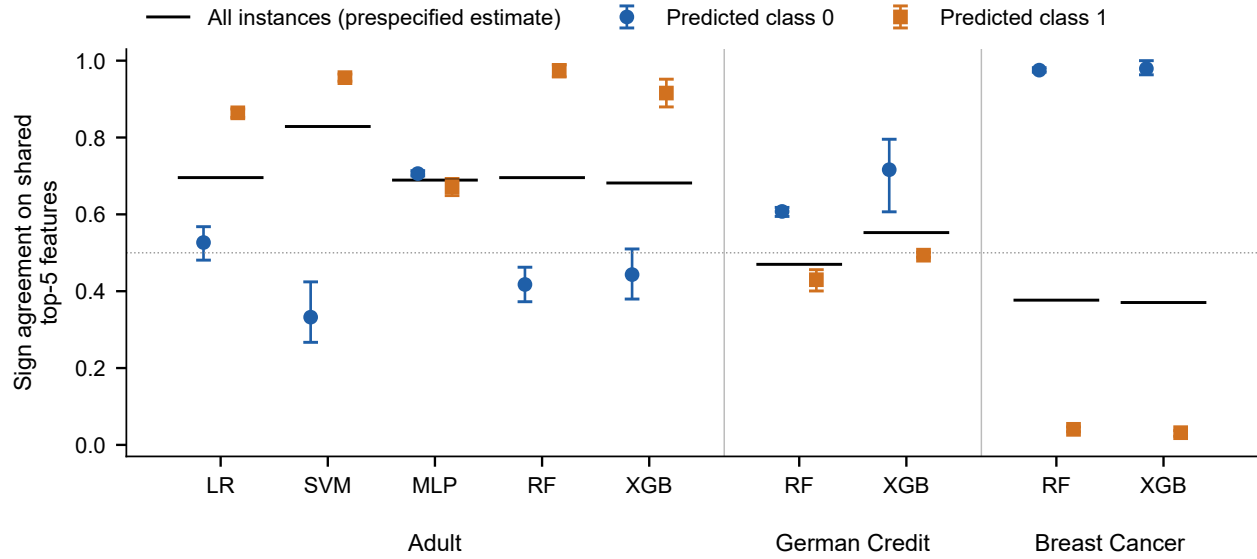


Fig. 2. Sign agreement between SHAP and LIME by predicted class, with seed-clustered 95% intervals. The dotted line marks one half

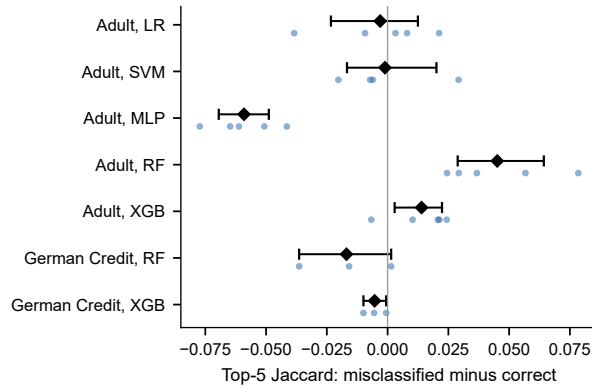


Fig. 3. Difference in top-5 overlap, misclassified minus correct. Points are seeds; diamonds are means with seed-clustered 95% intervals

Table 4. Mean within-run Spearman correlation between disagreement ($1 - J_5$) and explanation quality. Fid.: fidelity; Stab.: stability

Dataset	Model	SHAP		LIME	
		Fid.	Stab.	Fid.	Stab.
Adult	LR	-0.01	0.09	-0.40	0.12
	SVM	-0.05	0.06	-0.45	0.12
	MLP	-0.06	-0.24	-0.19	-0.07
	RF	-0.22	0.23	-0.27	0.10
	XGB	-0.15	0.14	-0.40	0.11
German Credit	RF	-0.05	-0.06	-0.26	-0.10
	XGB	-0.04	0.02	-0.24	-0.15
Breast Cancer	RF	-0.06	-0.56	-0.28	-0.54
	XGB	0.22	-0.34	0.01	-0.27

4.6 Feature Sets of Anchors and DiCE (Secondary)

Table 5 gives the overlap between feature sets on Adult. The features in an Anchors rule overlapped more with the SHAP top-5 set (0.348) than with the LIME top-5 set (0.240). The features changed by DiCE overlapped little with any other set (at most 0.078). These methods answer different questions,

a sufficient condition and a minimal change, so a low overlap is not an error of either method; it shows that their features cannot be substituted for those of an attribution method.

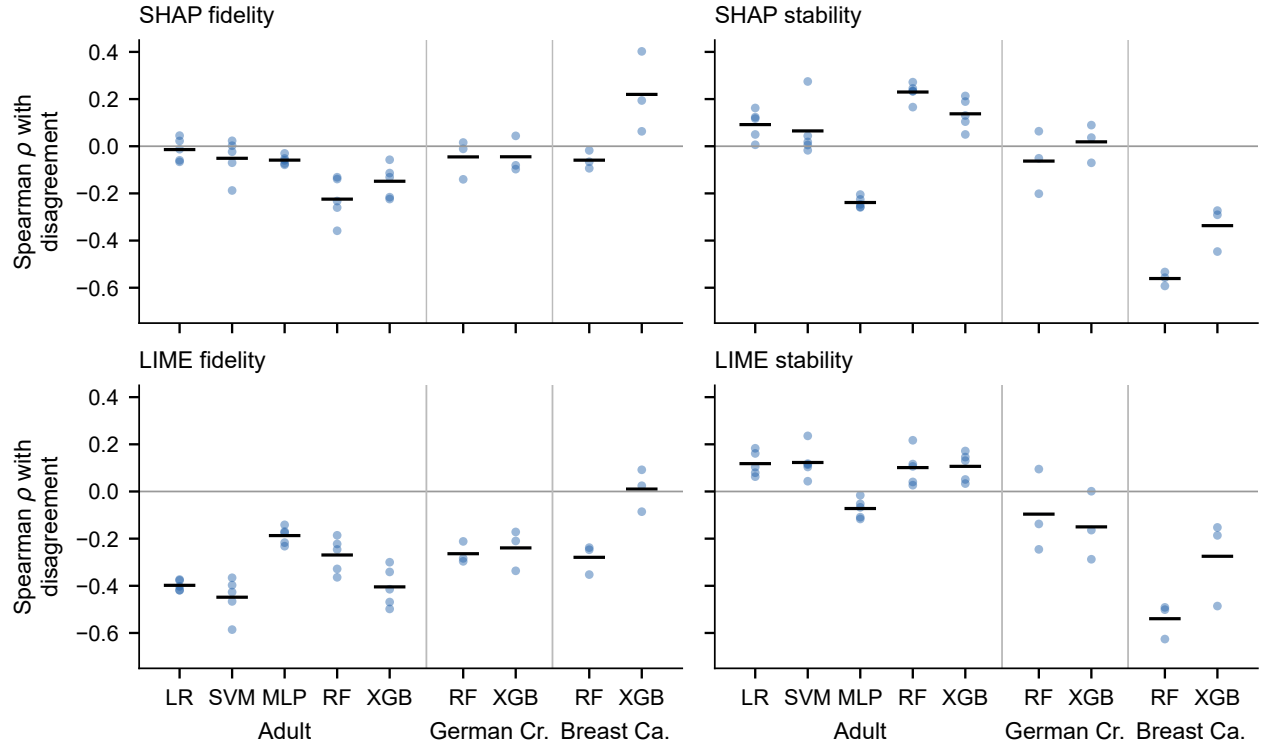


Fig. 4. Within-run Spearman correlation between disagreement and four quality values. Points are seeds; bars are means over seeds

Table 5. Jaccard overlap between feature sets on Adult (SHAP, LIME: top-5 features; Anchors: rule features; DiCE: changed features): mean of run means and seed-clustered 95% interval

Feature sets	Runs	Pairs	Mean	95% int.
SHAP vs. Anchors	56	14,584	0.348	[0.340, 0.356]
LIME vs. Anchors	57	19,095	0.240	[0.234, 0.247]
SHAP vs. DiCE	67	25,248	0.078	[0.068, 0.087]
LIME vs. DiCE	68	29,830	0.078	[0.074, 0.085]
Anchors vs. DiCE	53	18,578	0.068	[0.062, 0.075]

5 Discussion

The choice of explainer changes the explanation. On two of the three datasets, SHAP and LIME shared on average fewer than three of their five most important features, and the order of the features was almost unrelated for two model families. A user who reads a top-5 list would see a

different list with the other method. This agrees with the disagreement reported by Krishna et al. [14] and extends it with the run as the unit of analysis: the result is stable across five seeds and three sampling intensities, so it is not an artefact of a particular sample of instances.

Agreement is a property of the model and the data. The differences between model families and between datasets were larger than every other effect measured here. This is consistent with the view that the methods approximate the model in different neighbourhoods [9]: where the model is close to linear around the instance, as for logistic regression, the approximations coincide more. The high agreement on Breast Cancer indicates that low agreement is not inevitable. A report of explainer agreement should therefore state the model family and the dataset, and should not be generalised from one of them.

Signs need the same quantity. The sign analysis shows a risk for any study that compares the direction of attributions between methods. A LIME coefficient without discretisation is a slope; a SHAP value is a contribution relative to a reference [8, 16]. Their signs can differ for every instance of one class while the two methods describe the same relation between the feature and the output. Before signs are compared, the two explanations should be brought to the same quantity, for example by multiplying the slope by the deviation of the feature from the reference, or by using LIME with discretisation, whose weights refer to the value of the instance. The measures of overlap and rank used for RQ1 to RQ4 depend on absolute values and are not affected by this difference of sign, although a slope and a contribution also differ in magnitude, which is one source of the low rank concordance.

Disagreement and correctness. Agreement was not consistently lower when the model was wrong. The direction depended on the model family and the differences were small. Disagreement between explainers is therefore not a usable signal of a wrong prediction in these data.

Disagreement and LIME fidelity. Disagreement was higher where LIME fitted the model less well. Part of the disagreement may thus come from the approximation error of the local linear model, not from two valid but different views of the model. The fidelity of LIME, which is available at no extra cost, can indicate the instances for which the two methods are more likely to differ.

Limitations. The study covers three tabular binary datasets and one configuration of each explainer; other kernel widths, sample sizes or discretisation settings of LIME may change the estimates. There are five seeds on Adult and three on the other datasets, so the intervals are imprecise and the analyses of RQ2 to RQ4 are exploratory. Only the ten largest features of each explanation were stored, which limits the rank measure to that list. One-hot columns were treated as separate features; an analysis by original variable would give higher overlap. TreeSHAP was used with its default output, which for gradient boosting is the log-odds and for the random forest the probability, while LIME explains the probability; this affects

magnitudes and ranks, not signs. KernelSHAP failed for part of the SVM instances, so that family has fewer pairs. The models of the two smaller datasets were regenerated from the training recipe and validated against stored metrics and predictions, not compared byte by byte with the original binaries. For the Adult random forest, the stored binary does not reproduce all the predictions recorded in the runs; the analysis uses only the recorded outputs, and the paired SHAP and LIME records agree on every label and prediction. The sign analysis is post hoc.

6 Conclusions and Future Work

For the same instance and model, SHAP and LIME agreed only partly on the most important features: the mean top-5 overlap was 0.393 on Adult, 0.387 on German Credit and 0.714 on Breast Cancer. Agreement depended on the model family and the dataset, not on the number of explained instances, and it was not consistently lower for misclassified instances. Disagreement was associated with a lower local fidelity of LIME. The sign agreement between SHAP and LIME without discretisation followed the predicted class, because one method returns slopes and the other contributions; it should not be used as a measure of disagreement unless both explanations are first expressed as the same quantity.

Two practical consequences follow. An explanation shown to a user should name the explainer, since another explainer would often show other features. And a comparison of explainers should check, before any measure is computed, what each stored value means.

Future work will convert LIME coefficients into contributions to measure sign agreement on a common quantity, repeat the analysis by original variable, vary the LIME configuration, and add datasets to separate the effect of the number and type of features from that of the model family.

Data Availability

The three datasets are public [2, 11, 24]. The code, the per-instance run outputs, the new LIME cohort, the registered analysis plan and every result file are available at <https://github.com/jonnabio/xai-eval-framework>. Every number, table and figure in this article is generated from those files by the scripts described in `docs/reports/paper_e/README.md`.

Declaration on the Use of Artificial Intelligence

The author used the AI tool Claude (Anthropic), under the author's direction, to help write analysis and figure code, to draft and edit the English text and to format the references. The research questions, the analysis plan and the interpretation are the author's, and the author reviewed all content for accuracy.

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